

PROJECT REPORT

Electricity Consumption Forecasting Using Machine Learning

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ABSTRACT

This project develops a machine-learning-based system for electricity consumption forecasting. The supplied electricity dataset contains 48,048 records and 17 original columns covering electricity demand, weather variables, calendar information, holidays, and school indicators. The data was cleaned, converted to a proper datetime format, sorted chronologically, and transformed using time-based, lag, and rolling-window features.

After feature engineering and removal of the initial rows containing lag/rolling-window missing values, 47,880 records with 26 predictive features were used. The data was divided chronologically into 38,304 training records and 9,576 testing records. Linear Regression was used as a baseline, while Random Forest Regressor was developed as the main forecasting model.

On the test data, the Random Forest model produced an MAE of 19.36, RMSE of 27.35, and R^2 of 0.9784. The notebook also includes actual-versus-predicted plots, feature-importance analysis, residual analysis, forecast generation, and saving/loading of the trained model.

Keywords: Electricity Demand, Forecasting, Machine Learning, Random Forest, Linear Regression, Feature Engineering, Lag Features, Time-Series Data

1. INTRODUCTION

Electricity consumption changes over time because of daily usage patterns, seasonal effects, weather conditions, holidays, weekends, and other calendar factors. Accurate demand forecasting can support planning and operational decision-making by providing estimates of expected electricity demand.

This project uses historical electricity-demand observations together with weather and calendar-related variables to build a supervised machine-learning forecasting pipeline. Lag and rolling features are created so that previous demand values become explicit predictors.

The implementation is contained in a Google Colab notebook and uses Python libraries including pandas, NumPy, Matplotlib, scikit-learn, and joblib.

2. PROBLEM STATEMENT

The objective is to develop a machine-learning system that can learn patterns in historical electricity consumption and use engineered time, weather, calendar, lag, and rolling features to estimate electricity demand on unseen test observations.

3. OBJECTIVES

- Load and inspect the electricity consumption dataset.
- Convert the datetime field and organize observations chronologically.
- Create time, calendar, lag, and rolling features.
- Split the data chronologically into training and testing subsets.
- Develop Linear Regression as a baseline and Random Forest as the main model.
- Evaluate predictions using MAE, RMSE, and R^2 .
- Analyze feature importance and residuals.

- Save and reload the trained model and feature information.
- Generate and save forecast results.

4. DATASET DESCRIPTION

The notebook loads continuous dataset.csv. The original dataset contains 48,048 rows and 17 columns and covers 2015-01-03 01:00:00 through 2020-06-27 00:00:00. No missing values or duplicate rows were found in the original dataset.

Category	Fields
Target	nat_demand
Datetime	datetime
Weather – Toc	T2M_toc, QV2M_toc, TQL_toc, W2M_toc
Weather – San	T2M_san, QV2M_san, TQL_san, W2M_san
Weather – Dav	T2M_dav, QV2M_dav, TQL_dav, W2M_dav
Calendar / events	Holiday_ID, holiday, school

Target statistics: mean 1182.87, standard deviation 192.07, minimum 85.19, maximum 1754.88.

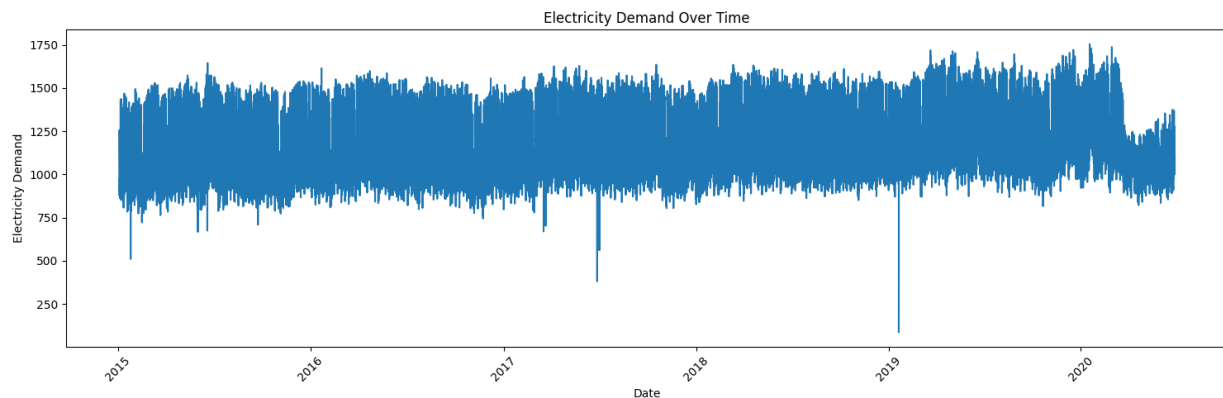


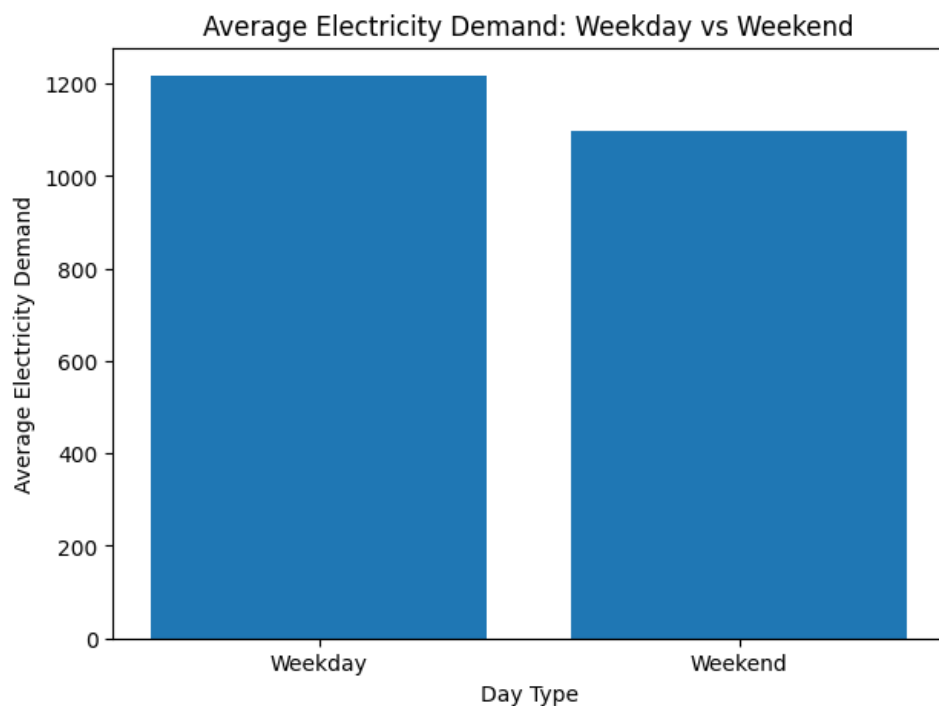
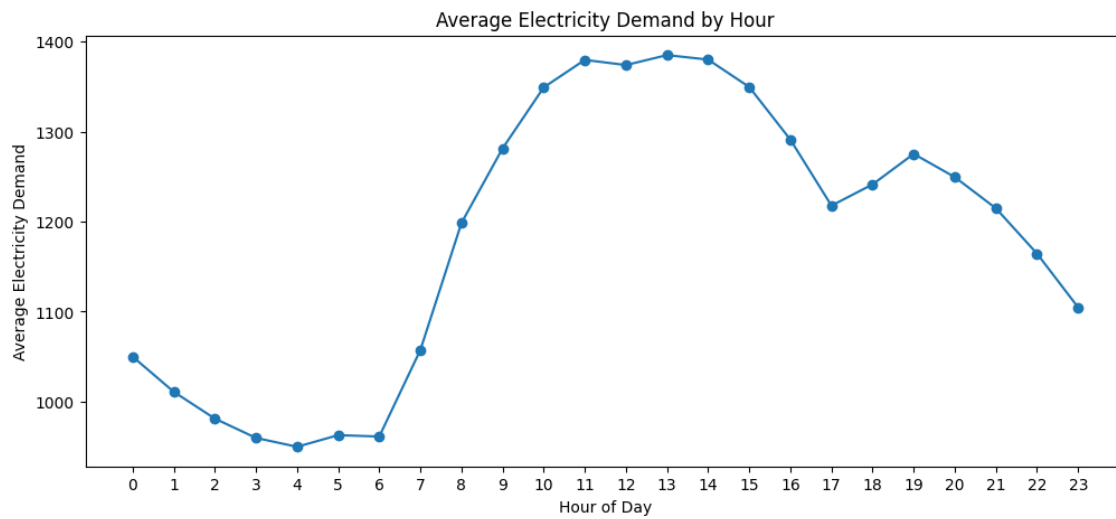
Figure 1. Electricity demand over time.

5. METHODOLOGY

Stage	Processing
Data collection	Load CSV dataset from Google Drive.
Data inspection	Check dimensions, columns, data types, duplicates, missing values.
Data preparation	Convert datetime and sort chronologically.
Time features	Year, month, day, hour, day_of_week, weekend.
EDA	Analyze demand, hourly/monthly patterns, day type, weather correlation.
Temporal features	Lag features and rolling means.
Split	80/20 chronological train/test split.
Modeling	Linear Regression and Random Forest.
Evaluation	MAE, RMSE, R^2 .
Analysis	Prediction curves, feature importance, residuals.
Persistence	Save and reload model/metadata.
Forecast	Generate 24 forecast points and save CSV.

6. EXPLORATORY DATA ANALYSIS

The notebook studies electricity-demand behavior over time, by hour, by weekday/weekend status, and in relation to weather variables.



The notebook reports average demand of 1216.84 for weekdays and 1097.94 for weekends. Hourly demand increases from the early morning into the daytime and then decreases later in the day.

7. FEATURE ENGINEERING

7.1 Time-Based Features

The notebook creates year, month, day, hour, day_of_week, and weekend.

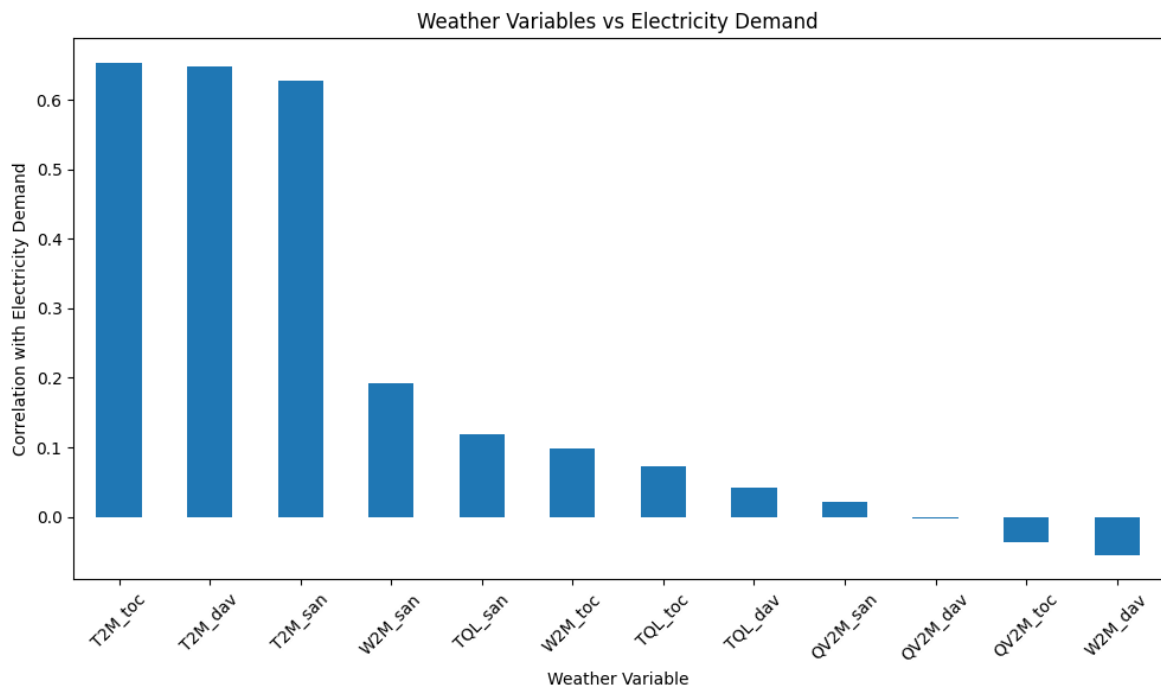
7.2 Lag Features

Feature	Meaning
demand_lag_1	Previous hourly demand.
demand_lag_24	Demand approximately 24 hours earlier.
demand_lag_168	Demand approximately 168 hours earlier.

7.3 Rolling Features

Feature	Meaning
rolling_mean_24	Mean of previous 24 demand observations.
rolling_mean_168	Mean of previous 168 demand observations.

After dropping the initial rows made incomplete by these temporal features, the final modeling dataset contains 47,880 rows and 26 features.



7.4 Final Feature Set

The 26 features consist of weather variables, event indicators, calendar variables, lag variables, and rolling-demand features.

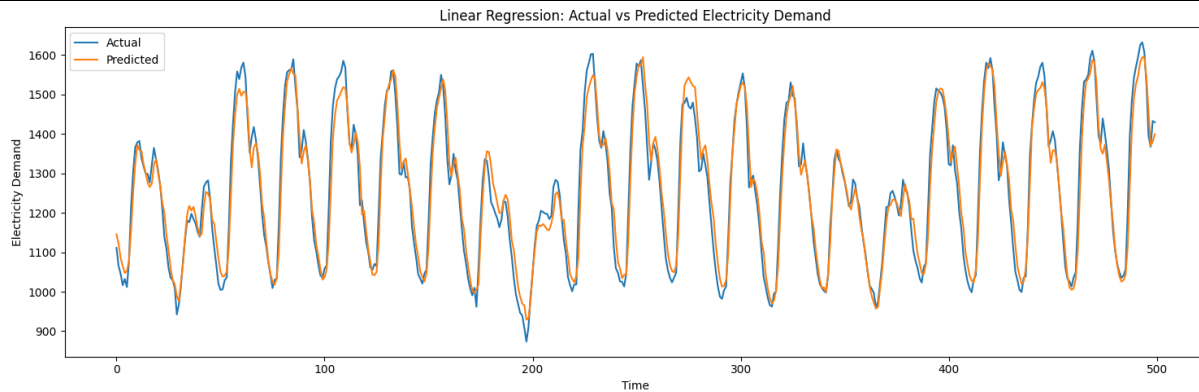
8. MODEL DEVELOPMENT

8.1 Linear Regression Baseline

Linear Regression was trained on standardized features as a baseline.

Metric	Linear Regression
MAE	37.02

RMSE	46.84
R^2	0.9366



8.2 Random Forest Regressor

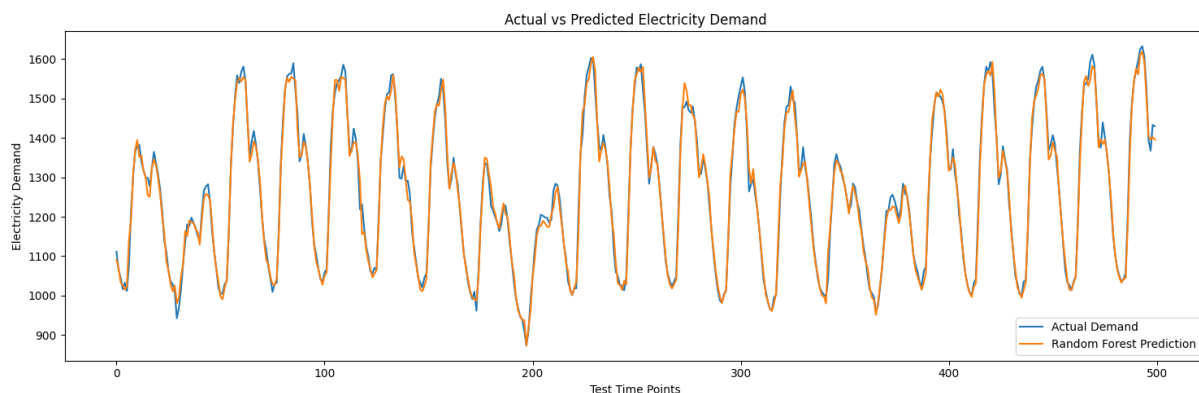
The main model is RandomForestRegressor with 100 estimators, random_state=42, and n_jobs=-1. It is trained on the engineered feature matrix.

Parameter	Value
Estimator	RandomForestRegressor
Number of estimators	100
Random state	42
n_jobs	-1

9. RESULTS AND DISCUSSION

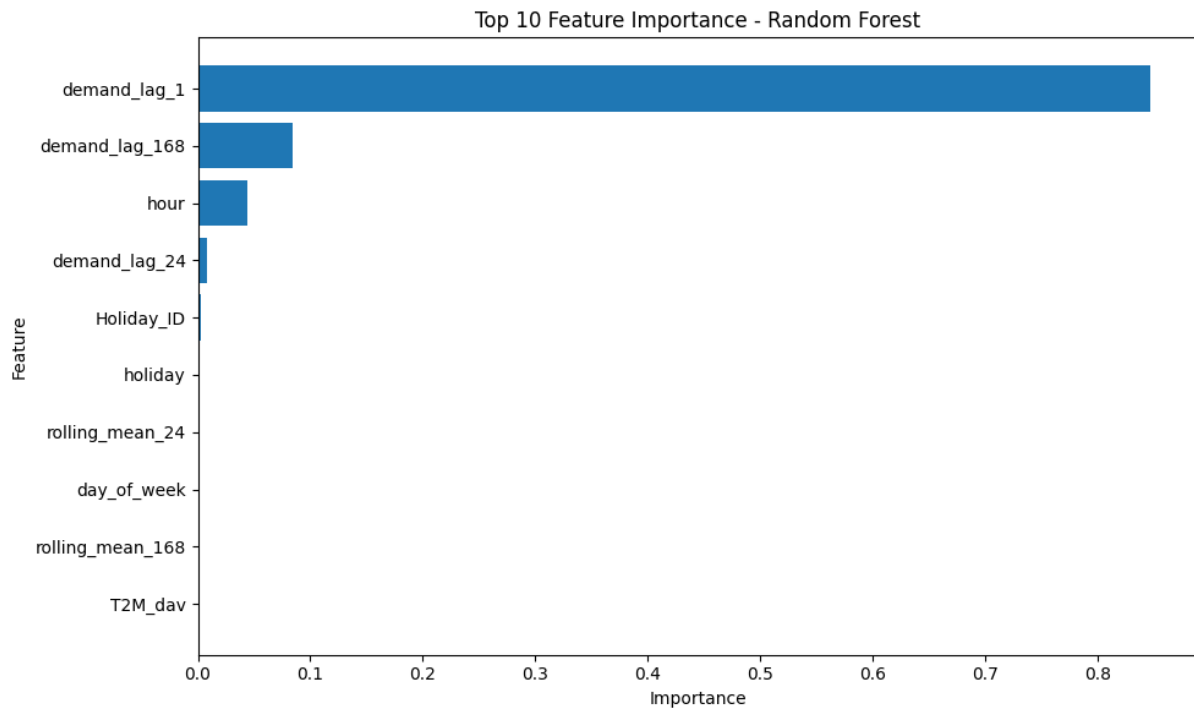
Model	MAE	RMSE	R^2
Linear Regression	37.02	46.84	0.9366
Random Forest	19.36	27.35	0.9784

On the test set recorded in the notebook, Random Forest produced lower MAE and RMSE and a higher R^2 than the Linear Regression baseline.



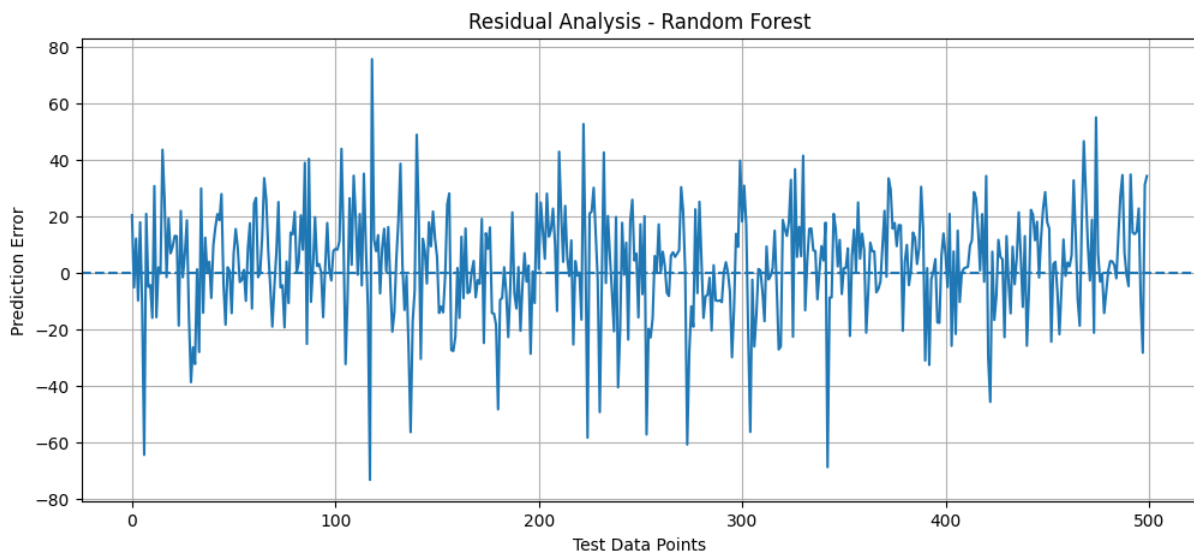
9.1 Feature Importance

The feature-importance analysis identifies demand_lag_1 as the dominant feature, followed by demand_lag_168 and hour among the most notable features.



9.2 Residual Analysis

Residuals are calculated as actual demand minus predicted demand. The residual plot is used to inspect the sequence of prediction errors.



9.3 Final Random Forest Metrics

Metric	Result
MAE	19.36296
RMSE	27.35158
R ² Score	0.97838

The notebook verifies the saved model by loading it again and generating predictions.

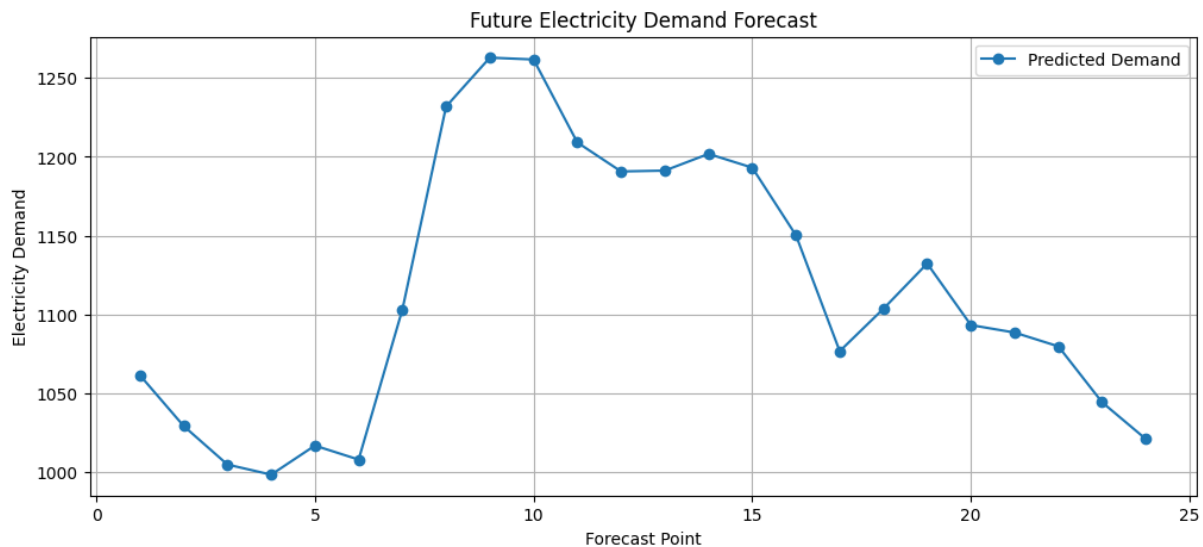
10. MODEL PERSISTENCE AND FORECAST GENERATION

The trained Random Forest model is saved using joblib, and feature names are stored as JSON metadata. The notebook reloads the model and uses it to generate predictions.

Artifact	Purpose
electricity_random_forest_model.pkl	Saved Random Forest model.
feature_scaler.pkl	Saved StandardScaler.
feature_names.json	Saved feature-name metadata.
future_electricity_forecast.csv	Forecast output table.

10.1 Forecast Generation

The notebook generates 24 forecast points from the selected test-feature subset and stores the resulting Forecast_Point and Predicted_Electricity_Demand columns in a CSV file.



11. LIMITATIONS

- A separate future-date feature-construction pipeline is needed for genuine forecasts beyond the available dataset.
- Only Linear Regression and Random Forest are evaluated in the current notebook.
- Weather correlation is descriptive and does not establish causation.
- The workflow is notebook-based and is not yet deployed as a continuously running service.

12. FUTURE SCOPE

- Develop a proper future-date forecasting pipeline.
- Compare XGBoost, LightGBM, LSTM, and other time-series methods.
- Integrate continuously updated weather data.
- Deploy the model through a web application or API.
- Build an interactive Power BI or Tableau dashboard.
- Extend forecasting to daily, weekly, and monthly horizons.
- Automate periodic retraining with new data.

13. CONCLUSION

The project implements an end-to-end electricity consumption forecasting workflow using machine learning. The dataset was inspected and prepared, time-based and temporal demand features were engineered, and a chronological train/test split was used.

Linear Regression produced MAE 37.02, RMSE 46.84, and R^2 0.9366. Random Forest produced MAE 19.36, RMSE 27.35, and R^2 0.9784 on the test data recorded in the notebook. Feature importance analysis showed that recent historical demand, particularly demand_lag_1, was highly influential.

The project also includes model persistence, actual-versus-predicted visualization, residual analysis, feature metadata storage, and forecast-output generation. It therefore provides a complete academic machine-learning workflow for studying and forecasting electricity demand.